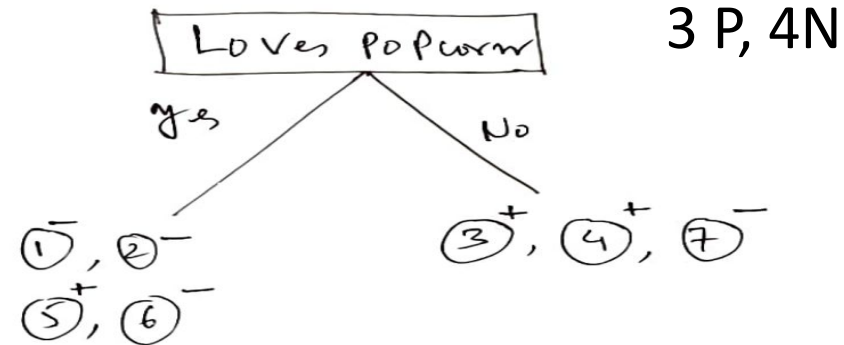


Step by Step Procedure for Decision Tree Construction

	Loves Popcorn	Loves Soda	Age	Loves Troll 2
①	Yes	Yes	7	No
②	Yes	No	12	No
③	No	Yes	18	Yes
④	No	Yes	35	Yes
⑤	Yes	Yes	38	Yes
⑥	Yes	No	50	No
⑦	No	No	83	No

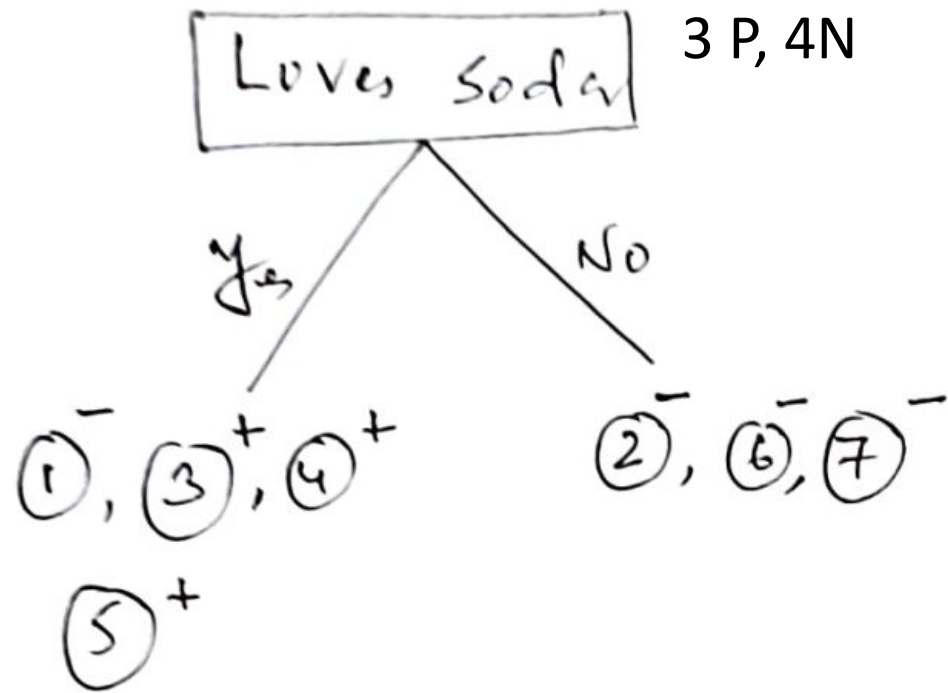


$$\begin{aligned}
 G(\text{Loves Popcorn}) &= \frac{4}{7} \left\{ 1 - \left(\frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2 \right\} \\
 &\quad + \frac{3}{7} \left\{ 1 - \left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)^2 \right\} \\
 &= 0.405
 \end{aligned}$$

$$\text{Info}(D) = \text{Info}(3P,4N) = - (3/7) \log_2(3/7) - (4/7) \log_2(4/7) = 0.985$$

$$\begin{aligned} \text{Info}(\text{Loves Popcorn}) &= (4/7) [-1/4 \log_2(1/4) - 3/4 \log_2(3/4)] \\ &\quad + (3/7) [-2/3 \log_2(2/3) - 1/3 \log_2(1/3)] \\ &= 0.857 \end{aligned}$$

$$\text{Info Gain}(\text{Loves Popcorn}) = 0.985 - 0.857 = 0.128$$



$G(\text{Loves Soda})$

$$= \frac{4}{7} \left\{ 1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2 \right\} + \frac{3}{7} \left\{ \left(\frac{0}{3}\right)^2 + \left(\frac{3}{3}\right)^2 \right\}$$

$$= 0.214$$

$$\text{Info}(D) = \text{Info}(3P,4N) = - (3/7) \log_2(3/7) - (4/7) \log_2(4/7) = 0.985$$

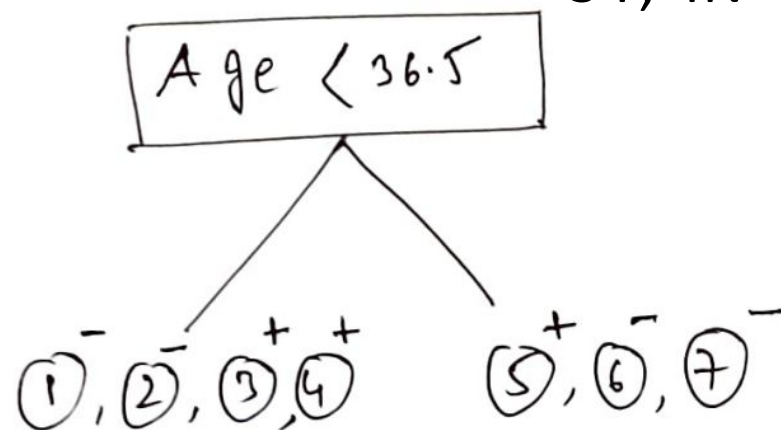
$$\begin{aligned} \text{Info}(\text{Loves Soda}) &= (4/7) [-3/4 \log_2(3/4) - 1/4 \log_2(1/4)] \\ &\quad + (3/7) [-0/3 \log_2(0/3) - 3/3 \log_2(3/3)] \\ &= 0.464 \end{aligned}$$

$$\text{Info Gain}(\text{Loves Soda}) = 0.985 - 0.464 = 0.521$$

Age

7 12 18 35 38 50 83
 ↓ ↓ ↓ ↓ ↓ ↓
 9.5 15 26.5 36.5 44 66.5

3 P, 4N



$G(\text{Age} < 36.5)$

$$= \frac{4}{7} \left\{ 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2 \right\} + \frac{3}{7} \left\{ 1 - \left(\frac{1}{3}\right)^2 - \left(\frac{2}{3}\right)^2 \right\}$$

$$= 0.476$$

$$\text{Info}(D) = \text{Info}(3P,4N) = - (3/7) \log_2(3/7) - (4/7) \log_2(4/7) = 0.985$$

$$\begin{aligned} \text{Info}(\text{Age} < 36.5) &= (4/7) [-2/4 \log_2(2/4) - 2/4 \log_2(2/4)] \\ &\quad + (3/7) [-1/3 \log_2(1/3) - 2/3 \log_2(2/3)] \\ &= 0.965 \end{aligned}$$

$$\text{Info Gain}(\text{Age} < 36.5) = 0.985 - 0.965 = 0.02$$

Age < 9.5
0.429

Age < 15
0.343

Age < 26.5
0.476

Age < 36.5
0.476

Age < 44
0.343

Age < 66.5
0.429

Age < 15 / Age < 44 → Winner

Now

$$G(\text{Loves Soda}) = 0.214$$

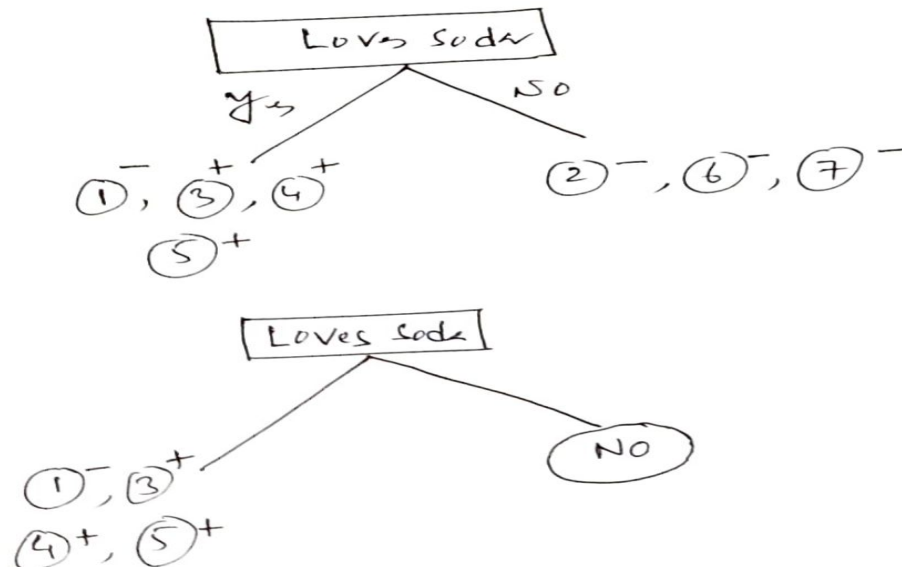
$$G(\text{Loves Popcorn}) = 0.405$$

$$G(\text{Age} < 15) = 0.343$$

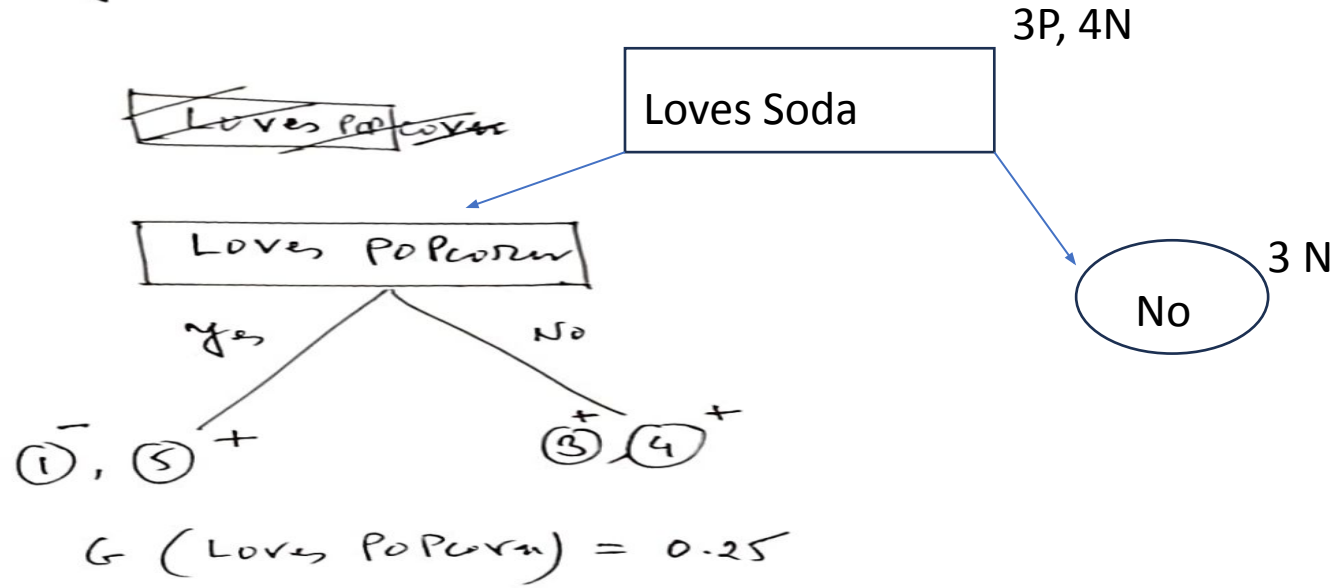
$$\text{Info Gain}(\text{Loves Popcorn}) = 0.128$$

$$\text{Info Gain}(\text{Loves Soda}) = 0.521$$

$$\text{Info Gain}(\text{Age} < 15) = 0.291$$



	Loves Popcorn	Loves Soda	Age	Loves Troll 2
①	Yes	Yes	7	No
③	No	Yes	18	Yes
④	No	Yes	35	Yes
⑤	Yes	Yes	38	Yes



Age

7 18 35 38

↓ ↓ ↓

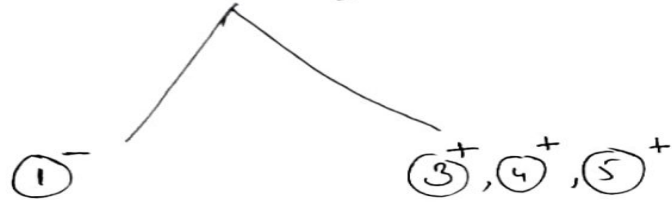
12.5 26.5 36.5

$$\text{Info}(D) = \text{Info}(3P,1N) = - (3/4) \log_2(3/4) - (1/4) \log_2(1/4) = 0.811$$

$$\begin{aligned} \text{Info}(\text{Loves Popcorn}) &= (2/4) [-1/2 \log_2(1/2) - 1/2 \log_2(1/2)] \\ &\quad + (2/4) [-2/2 \log_2(2/2) - 0/2 \log_2(0/2)] \\ &= 0.50 \end{aligned}$$

$$\text{Info Gain}(\text{Loves Popcorn}) = 0.811 - 0.50 = 0.311$$

$$G(\text{Age} < 12.5)$$



$$\begin{aligned}
 G(\text{Age} < 12.5) &= \frac{1}{4} \left\{ 1 - \left(\frac{0}{1}\right)^2 - \left(\frac{1}{1}\right)^2 \right\} \\
 &\quad + \frac{3}{4} \left\{ 1 - \left(\frac{3}{3}\right)^2 - \left(\frac{0}{3}\right)^2 \right\} \\
 &= 0
 \end{aligned}$$

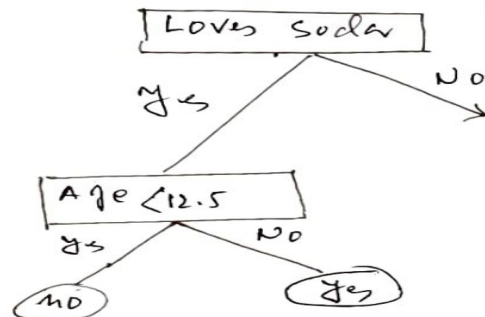
Calculate $G(\text{Age} < 26.5)$ and $G(\text{Age} < 36.5)$

$G(\text{Age} < 12.5) = 0$ is the minimum

$$G(\text{Loves Popcorn}) = 0.25$$

$$G(\text{Age} < 12.5) = 0$$

$G(\text{Age} < 12.5)$ is the winner



$$\text{Info}(D) = \text{Info}(3P,1N) = - (3/4) \log_2(3/4) - (1/4) \log_2(1/4) = 0.811$$

$$\begin{aligned}\text{Info}(\text{Age}<12.5) &= (1/4) [-0/1 \log_2(0/1) - 1/1 \log_2(1/1)] \\ &\quad + (3/4) [-3/3 \log_2(3/3) - 0/3 \log_2(0/3)] \\ &= 0.00\end{aligned}$$

$$\text{Info Gain}(\text{Age}<12.5) = 0.811 - 0.00 = 0.811$$

Info gain of other mid points are less than 0.811

$$\text{Info Gain}(\text{Loves Popcorn}) = 0.811 - 0.50 = 0.311$$

Age<12.5 is the
winner

To Avoid over fitting we may put a restriction on the number of examples in a ~~leaf~~^{node}. If a ~~leaf~~^{node} contains, i.e., 10 or less examples, this ~~leaf~~^{node} should not be extended further and converted to a leaf. The majority class in the leaf will determine the label of the leaf.

The cross validation approach may be used to find out the appropriate number of examples.

The other more effective approach for avoiding over fitting is tree pruning.